

NOR DAYANA ATHIRA BT MOHAMAD TOEL

Backend Software Engineer · Node.js & NestJS

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PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years of professional experience specialising in Node.js and NestJS, focused on designing and delivering scalable, secure and high-performance APIs. Proven experience in backend architecture, database design, authentication (JWT, Microsoft Entra ID/OAuth2), real-time WebSocket dashboards and IoT data processing. Comfortable owning backend features end-to-end, integrating external APIs (Hikvision, DroneDB, Billplz) and deploying production workloads via Docker, Kubernetes and CI/CD pipelines.

PROFESSIONAL EXPERIENCE

Backend Software Engineer | **TM Research & Development** - Cyberjaya

Mar 2022 – Present

API Development & Business Critical Systems

- Developed and maintained **NestJS REST APIs** for multiple enterprise applications: fleet management & vehicle tracking, workforce management & analytics dashboards, OSHE systems, and IDP platforms.
- Optimised the IDP platform backend** to support **5,000 concurrent users** with minimal errors through code-level refactoring, query optimisation and efficient resource management.
- Implemented secure user authentication using **Microsoft Entra ID (OAuth2/OIDC)** and **JWT tokens**, enabling SSO across 4+ enterprise applications and reducing unauthorised access incidents to zero.
- Built real-time WebSocket based dashboard features** for live data streaming and workforce analytics, serving concurrent users across multiple tenant organisations.
- Integrated **Hikvision external APIs** for camera and device data ingestion into dashboard and reporting, enabling live surveillance data feeds.
- Designed and implemented a backend data processing engine for IoT device data, ensuring accurate and reliable event capture at scale.
- Built **multi-tenant backend architecture** supporting clustered deployments across multiple organisations.
- Integrated **Billplz payment gateway** into a microservices-based system, enabling secure and reliable transaction processing.
- Integrated external APIs including DroneDB, handling geospatial and operational data.

Backend Infrastructure & DevOps

- Designed, deployed and maintained containerised backend services using **Docker** across development, staging and production environments.
- Automated microservices deployments via **GitLab CI/CD** integrated with Cloud Alpha Edge (CAE) and Kubix, improving release reliability and cutting deployment time significantly.
- Supported **Kubernetes based deployments**, gaining hands-on experience with container orchestration.
- Implemented and managed **RabbitMQ** for reliable inter-service communication, including secure data stream exposure to external partners.
- Configured and supported **Nginx** for backend service routing and environment-based deployments.
- Monitored server performance and application health to ensure high availability and prevent service downtime.

Cloud Migration & Data Reliability

- Led migration of systems from Digital Ocean to Cloud Alpha Edge, deploying applications using **Docker Compose** and setting up workflows with Node-RED.
- Maintained **PostgreSQL, MySQL and MongoDB** databases; performed scheduled backup and recovery operations ensuring data integrity across production systems.

Initiatives & R&D

- Took initiative to explore blockchain technology by deploying and configuring a **Hyperledger Besu node** on the CyberChain network, gaining hands-on experience with distributed ledger systems.

Backend Intern | **TM Research & Development** - Cyberjaya

Mar 2022 (Internship — absorbed full-time)

- Contributed to backend API development for mobile applications using Node.js and Express.
- Assisted in enhancing backend integrations and application functionality; gained exposure to Angular frontend.
- Completed structured training in Node.js and MySQL, worked within an Agile/Scrum development environment.

Backend Intern | **Epnex Technologies Sdn Bhd** - Bandar Sunway

Jul 2019 – Sep 2019

- Tested and documented API performance using Node.js, conducted research and data analysis for web development projects.
- Developed a web application prototype using HTML and JavaScript, supported the team on face recognition / person detection evaluation.

TECHNICAL SKILLS

Languages: Node.js, TypeScript, Python, SQL, Shell Script, NoSQL

Frameworks: NestJS, Express, Angular

Auth & Security: Microsoft Entra ID (OAuth2 / OIDC), JWT, Role-Based Access Control (RBAC)

Databases: PostgreSQL, MySQL, MariaDB, MongoDB

Messaging & Realtime: RabbitMQ, WebSocket

DevOps & Infra: Docker, Docker Compose, Kubernetes, Nginx, GitLab CI/CD, Node-RED

ORM / Libraries: MikroORM

External APIs: Hikvision, DroneDB, Billplz, Microsoft Graph

Tools: Git, Postman, Jira, Confluence, VS Code

EDUCATION

Bachelor of Computer Science (Hons) | **Universiti Teknologi MARA (UiTM)** - Tapah

2020 – 2022

Relevant coursework: Database Engineering, Algorithm Analysis & Design

Diploma in Computer Science | **Universiti Teknologi MARA (UiTM)** - Segamat

2017 – 2020

AWARDS & ACTIVITIES

- 3rd Place — Team Programming Competition (Javathon @ UiTM Segamat, 2019)
- Committee Member — Kelab Inspirasi, UiTM Segamat (2017–2018)